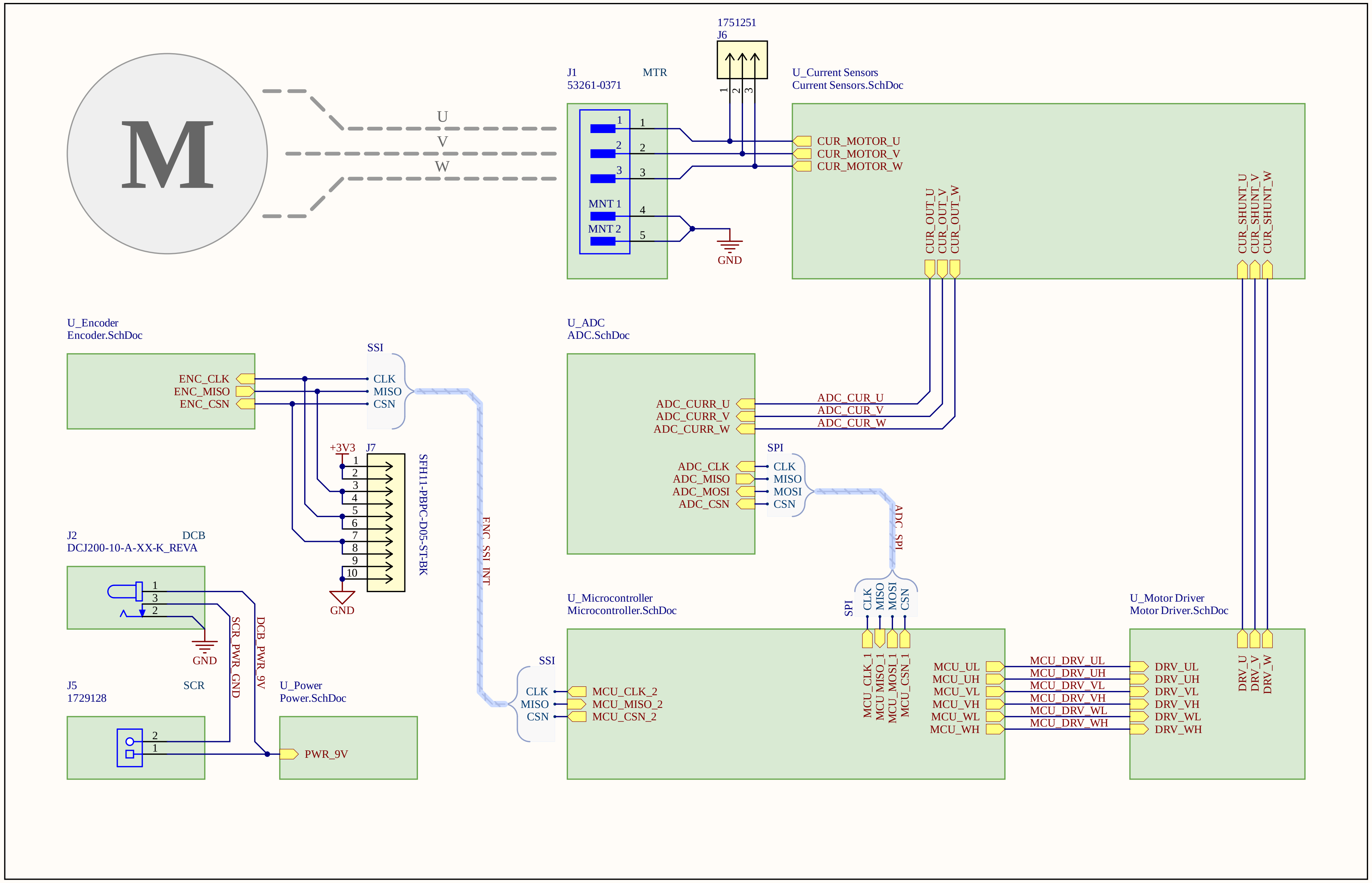
HAPTIC KNOB - FOR SIMULATING ELECTRICAL PHENOMENA

Description.

UBC Open Robotics
2025/2026



Patrick Delfin, Harry Nguyen, Trieu Truong

Legend:
MCU = Microcontroller
PWR = Power System
DRV = Motor Driver
CUR = Current Sensor
ADC = Analog to Digital Converter
ENC = Encoder
DCB = DC Barrel Jack
MTR = Motor Connector
SCR = Screw-In Terminal

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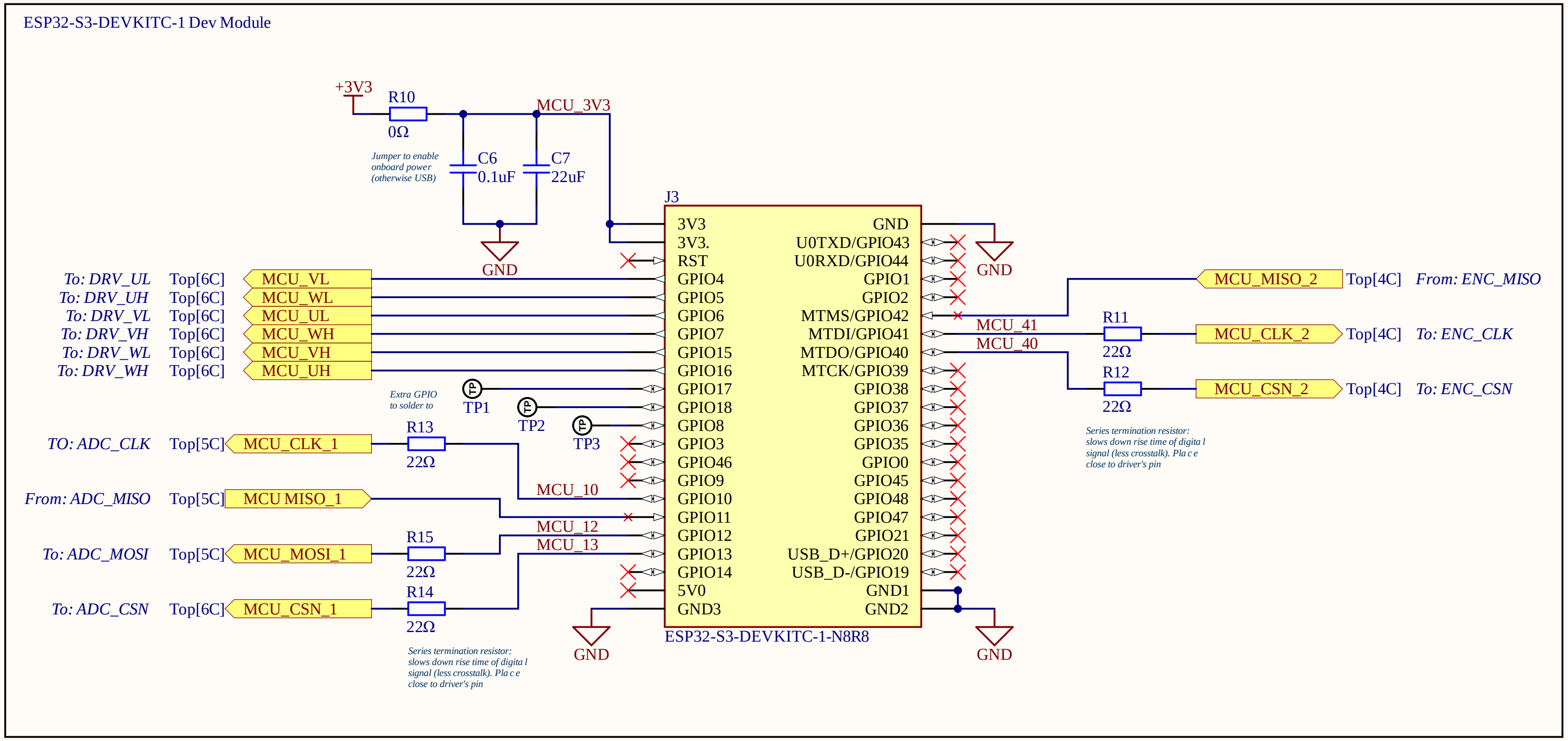
MICROCONTROLLER

Executes the control system including reading sensors, controlling the motor driver, and performing the haptics calculations.

Harry Nguyen
2026-02-12



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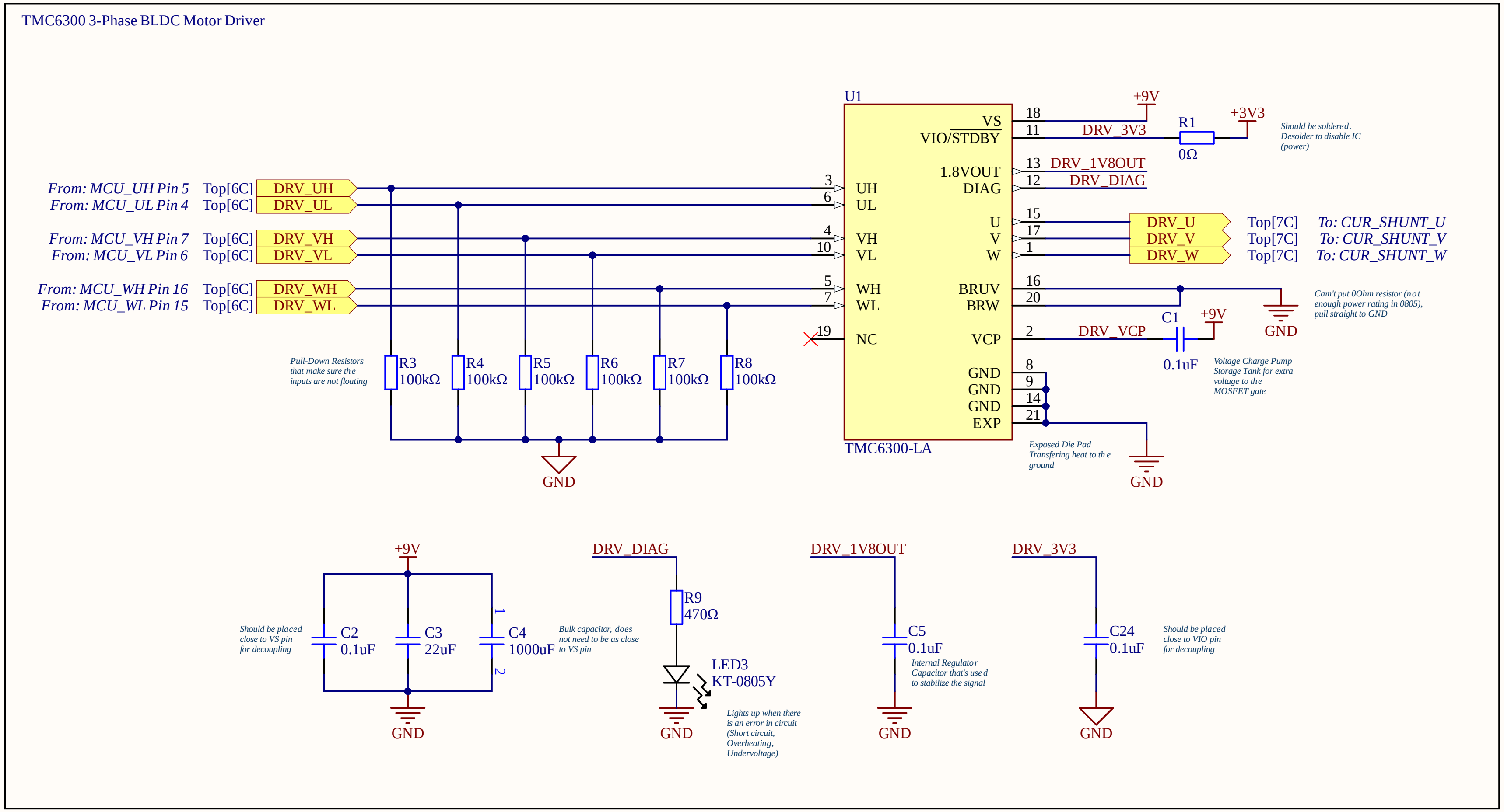
File: C:\Users\..\Microcontroller.SchDoc		Drawn By:

MOTOR DRIVER

Provides the necessary high current and voltage to manage speed, direction, and torque.

Trieu Truong
2026-01-31



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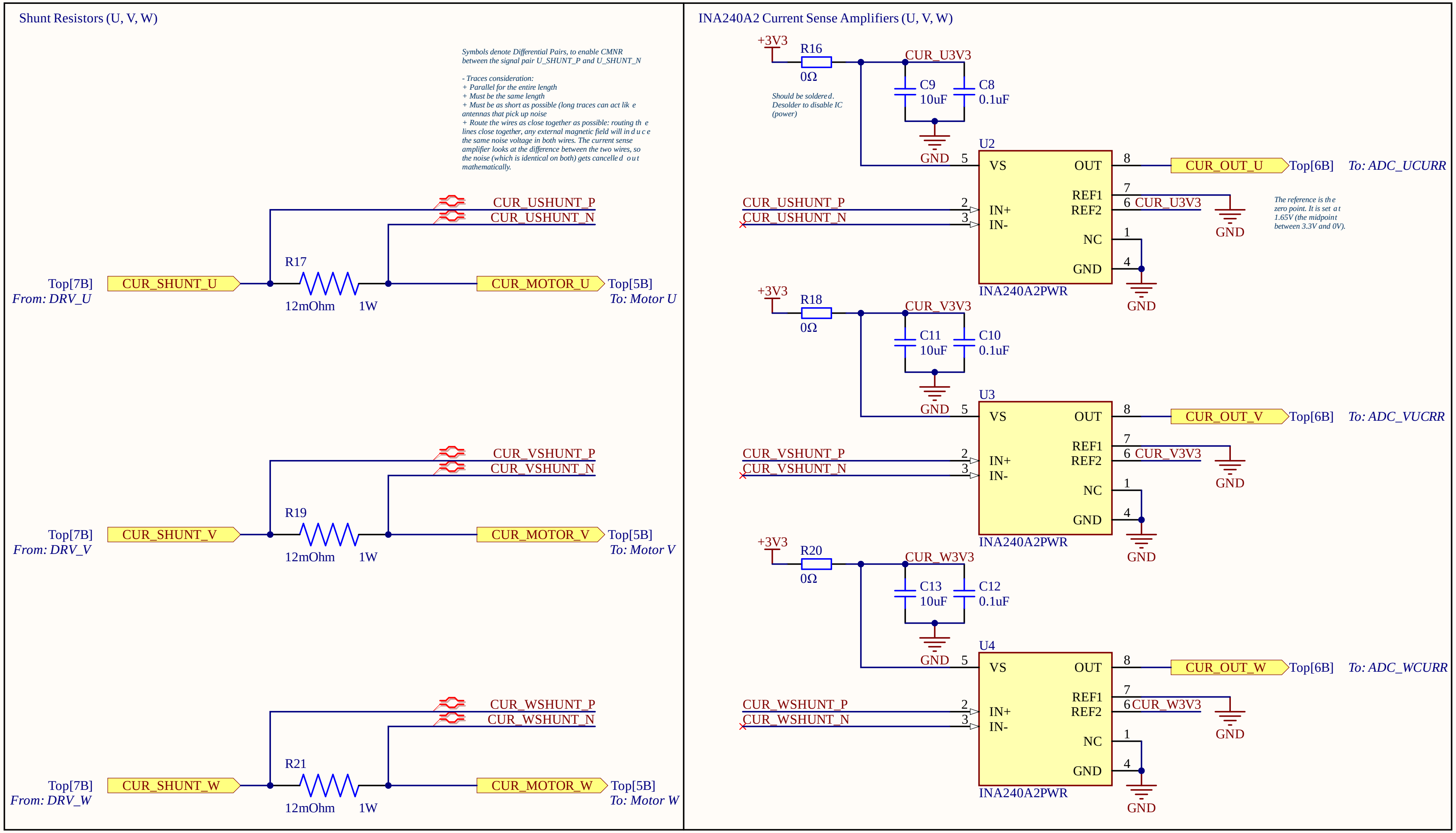
SENSOR - CURRENT SENSORS

Measures the current through each of the 3 phases by measuring the voltage drop across shunt resistors.

Harry Nguyen
2026-02-28



Notes:
INA240A2 Gain: 50V/V

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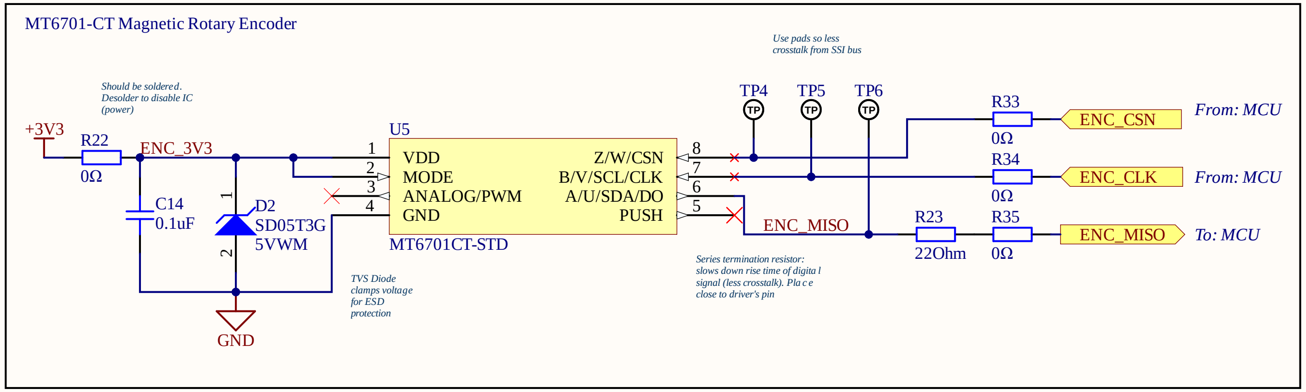
SENSOR - MAGNETIC ROTARY ENCODER

Reads the absolute angle of the motor's rotor.

Patrick Delfin
2026-01-23



Notes:
Configured Output: SSI (SPI)
Powered By: 3.3V linear voltage regulator
Max RPM: 55,000 RPM
Package: SOP-8

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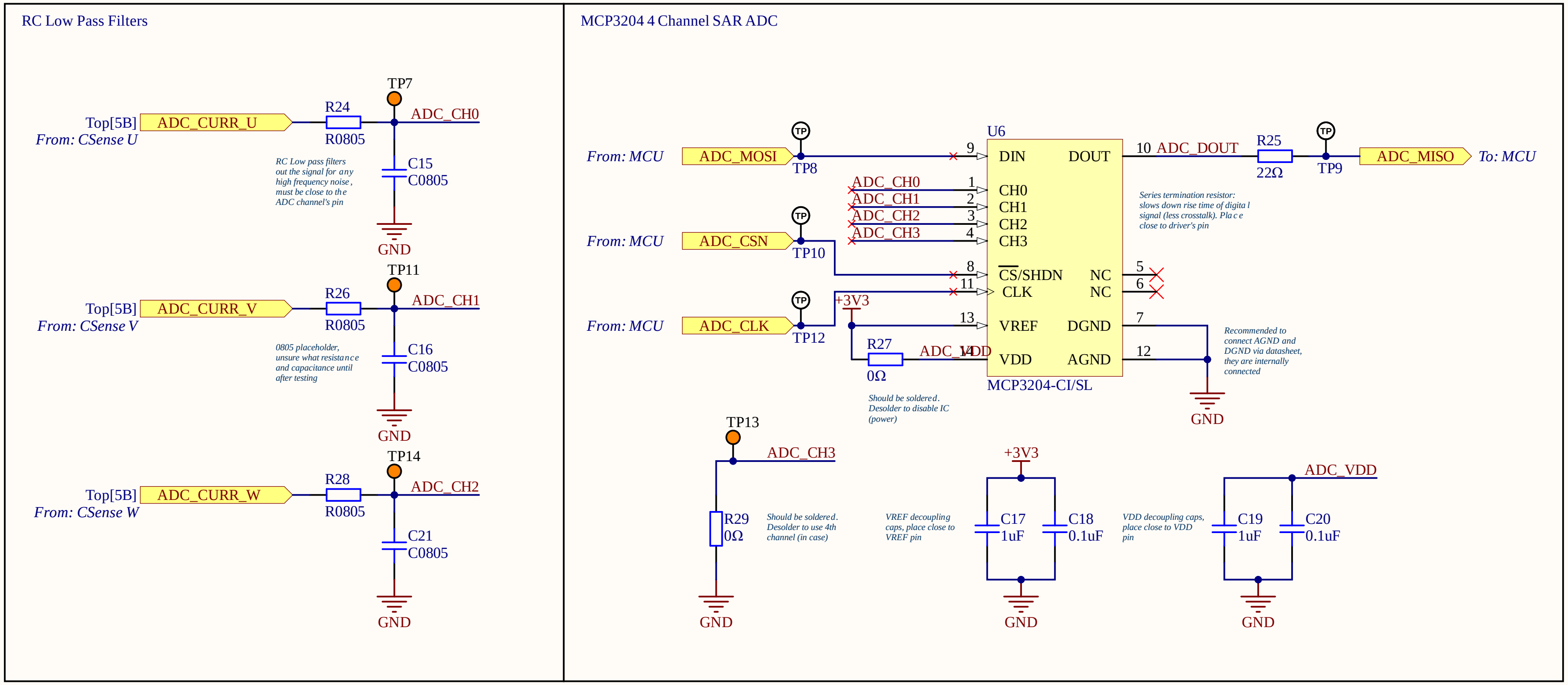
ANALOG TO DIGITAL CONVERTER

Converts analog current reading to digital SPI protocol.

Patrick Delfin
2026-02-04



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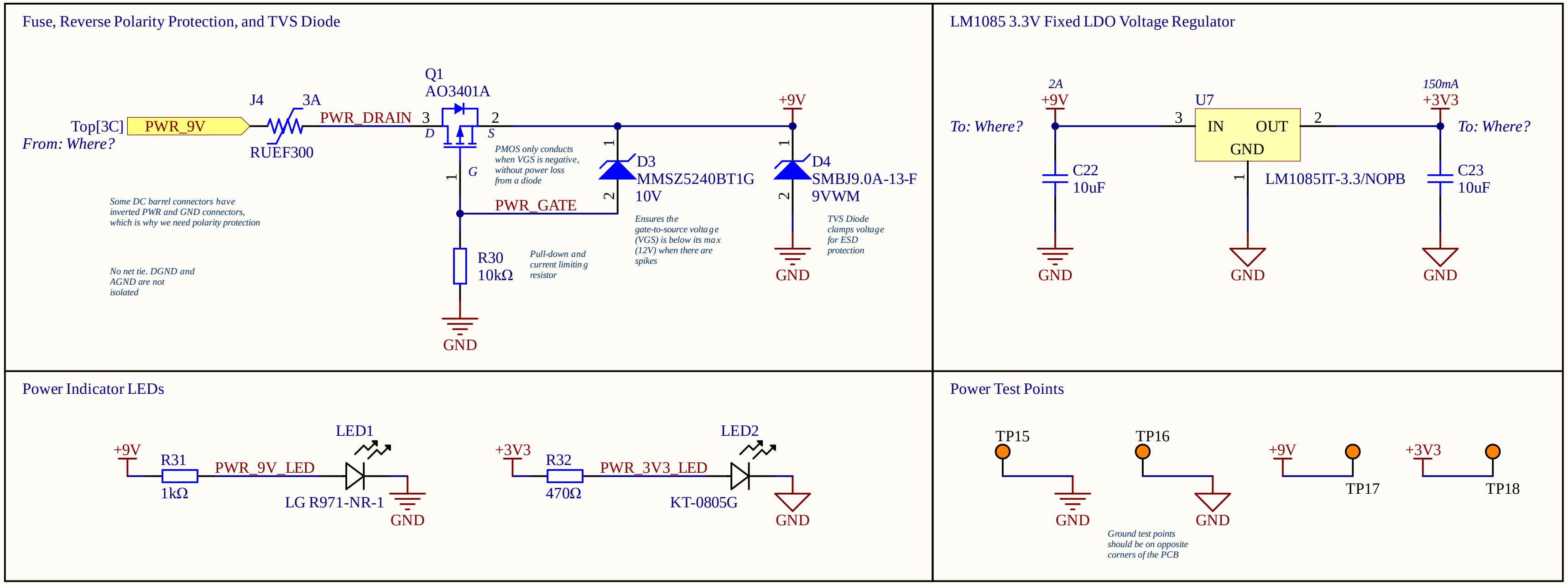
POWER SYSTEM

Supplies a protected 9V and 3.3V to the entire system.

Patrick Delfin
2026-02-12



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